Problem D
Double Up
Time limit: 5 seconds

A Double Up game consists of a sequence of n numbers $a_1, \ldots, a_n$, where each a_i is a power of two. In one move one can either remove one of the numbers, or merge two identical adjacent numbers into a single number of twice the value. For example, for sequence $4, 2, 2, 1, 8$, we can merge the 2s and obtain $4, 4, 1, 8$, then merge the 4s and obtain $8, 1, 8$, then remove the 1, and, finally, merge the 8s, obtaining a single final number, 16. We play the game until a single number remains. What is the largest number we can obtain?

Input

The input consists of two lines. The first line contains n ($1 \leq n \leq 1000$). The second line contains numbers $a_1, \ldots, a_n$, where $1 \leq a_i \leq 2^{100}$ for each i.

Output

The ouput consists of a single line containing the largest number that can be obtained from the input sequence $a_1, \ldots, a_n$.

Sample Input 1	Sample Output 1
5 4 2 2 1 8	16

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